

Example

Task

Find the weights of the single layer perceptron after one iteration.

- Inputs: $x_1=0.5$, $x_2=-0.2$, $x_3=0.15$
- Initial weights: $w_1=0.2$, $w_2=2$, $w_3=0.1$, $w_0=1$
- Target: $t=1$
- Learning speed: 0.1
- Transfer function– sigmoidal.

1. Forward pass: y, mse

$$mse = 1/2 (t - y)^2$$

$$net = \omega_0 + \sum_{k=1}^3 \omega_k x_k = 1 + 0.2 \times 0.5 + 2 \times (-0.2) + 0.1 \times 0.15 = 0.715$$

$$y = f(net) = \frac{1}{1 + e^{-net}} = 0.6715$$

$$mse = 1/2 (t - y)^2 = 0.5 (1 - 0.6715)^2 = 0.053$$

2. Backward pass:

$$\frac{\partial mse}{\partial \omega_k} = \frac{\partial mse}{\partial y} \frac{\partial y}{\partial net} \frac{\partial net}{\partial \omega_k}$$

$$\frac{\partial mse}{\partial y} = (y - t)$$

$$\frac{\partial y}{\partial net} = (1 - y)y$$

$$\frac{\partial net}{\partial \omega_1} = x_1$$

$$\frac{\partial mse}{\partial y} = (y - t) = (0.6715 - 1) = -0.328$$

$$\frac{\partial y}{\partial net} = (1 - y)y = 0.22$$

$$\frac{\partial net}{\partial \omega_1} = x_1 = 0.5$$

$$\frac{\partial mse}{\partial \omega_1} = \frac{\partial mse}{\partial y} \frac{\partial y}{\partial net} \frac{\partial net}{\partial \omega_1} = -0.328 \times 0.22 \times 0.5 = -0.036$$

Error of a neuron:

$$\delta_{output} = \frac{\partial mse}{\partial y} \frac{\partial y}{\partial net} = -0.328 \times 0.22 = -0.072$$

3. Weight w1 update

$$\omega_k \leftarrow \omega_k - \eta \frac{\partial mse}{\partial \omega_k}$$

$$\mathbf{w}_k \leftarrow \mathbf{w}_k - \eta \delta_{output} \mathbf{x}_k$$

$$\omega_1 = 0.2 - 0.1 (-0.072) 0.5 = 0.204$$